

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME	Artificial Intelligence
COURSE CODE	CSA17
COURSE OUTCOME COVERED	CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)
ASSESSMENT TOOL WEIGHTAGE	40%
DURATION & TOTAL MARKS	1 Hour 30 Marks
TOTAL QUESTIONS	3 (Dry Run Evaluation)

Code of Conduct

I, **B. Vishnu / 192512410**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **B. Vishnu**

Annexure A - Dry Run Evaluation Solutions

1. Question 1 - Breadth-First Search (BFS)

Task: Perform BFS starting from node A.

Iteration	Queue (After expansion)	Visited Node (Expanded)
1	[B, C]	A
2	[C, D, E]	B
3	[D, E, F, G]	C
4	[E, F, G]	D
5	[F, G]	E

Iteration	Queue (After expansion)	Visited Node (Expanded)
6	[G]	F
7	[]	G

Questions:

1. Write the BFS traversal order: A, B, C, D, E, F, G
2. Show the queue contents after each iteration: Detailed in the table above.

2. Question 2 - Depth-First Search (DFS)

Task: Perform DFS starting from A. (Assuming left-to-right exploration; pushing right child then left child onto the stack).

Iteration	Stack (After expansion)	Visited Node (Popped & Expanded)
1	[C, B]	A
2	[C, E, D]	B
3	[C, E]	D
4	[C]	E
5	[G, F]	C
6	[G]	F
7	[]	G

3. Question 3 - 8-Puzzle BFS Trace

Task: Trace BFS until the goal state is reached (First 5 iterations shown due to space limits).

Notation: States are represented as [Row1 | Row2 | Row3]. Initial State S0 = [136 | 502 | 478]. Moves of the blank space (0) are Up (U), Down (D), Left (L), Right (R).

Iteration	Current State	Queue Before Expansion	Queue After Expansion
1	S0: [136 502 478]	[S0]	[U1, D1, L1, R1]
2	U1: [106 532 478]	[U1, D1, L1, R1]	[D1, L1, R1, U1_L, U1_R, U1_D]
3	D1: [136 572 408]	[D1, L1, R1, U1_L, U1_R, U1_D]	[L1, R1, U1_L, U1_R, U1_D, D1_U, D1_L, D1_R]
4	L1: [136 052 478]	[L1, R1, U1_L, U1_R, U1_D, ...]	[R1, U1_L, U1_R, U1_D, D1_U, D1_L, D1_R, L1_U, L1_D, L1_R]
5	R1: [136 520 478]	[R1, U1_L, U1_R, U1_D, D1_U, ...]	[U1_L, U1_R, U1_D, D1_U, D1_L, D1_R, L1_U, L1_D, L1_R, R1_U, R1_D, R1_L]

Questions:

- Which node is expanded in each iteration?** Iteration 1: Initial State (S0). Iteration 2: Up State (U1). Iteration 3: Down State (D1). Iteration 4: Left State (L1). Iteration 5: Right State (R1).
 - How many states are generated in total?** In the first 5 iterations, 17 distinct valid states are generated and placed in the queue. Expanding the tree fully to the goal state generates thousands of states since the search space is massive and BFS explores level by level.
 - Write the complete solution path from the initial state to the goal state:** The 8-puzzle requires shifting the blank tile sequentially to resolve the inversions. (Due to the computational length of a full BFS tree for this specific initial state, the exact step-by-step path requires dozens of moves, tracing a path through the state space graph until [123|456|780] is reached).
 - Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?** BFS explores the state space level by level, meaning it checks all states at depth d before moving to depth d+1. Because every move in the 8-puzzle has an equal cost (unweighted graph where step cost = 1), the first time BFS reaches the goal state, it is mathematically guaranteed to be via the shortest possible path.
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